AI Shopping Assistant — Walkthrough

✅ Verified Working

Full end-to-end testing completed successfully:

Feature	Status	Details
Login	✅	user@shop.com / user123 — navbar shows "Demo User"
Products	✅	194 products loaded from DummyJSON with images, prices, ratings
Product Detail	✅	Specs, sentiment bar, reviews with sentiment tags
Add to Cart	✅	Cart badge updates instantly, "Added to cart!" message
Cart Page	✅	Shows item, quantity controls, order summary
Chatbot Widget	✅	Floating 🤖 button, NLP intent classification
Admin Dashboard	✅	Stats, sentiment overview, order/user management

Product Detail Page

Product detail page showing Amazon Echo Plus with specs and add-to-cart

Full Flow Recording

Full user flow: login → browse → add to cart → verify cart

What Was Built

🤖 AI Features (Self-Contained, No External APIs)

- **NLP Chatbot** — 14 intents (search, cart, orders, reviews, recommendations)
- **Sentiment Analysis** — Lexicon-based with negation/intensifier handling
- **Recommendation Engine** — Hybrid content-based + collaborative filtering

🛍️ Pages

- Home (hero + featured + categories + AI showcase)
- Products (search, filters, sort, pagination)
- Product Detail (specs, sentiment chart, review form)
- Cart (quantity controls, order summary, checkout)
- Dashboard (orders, AI recommendations, profile)
- Admin (stats, sentiment overview, order/user management)
- Login / Register

Project Structure

```
src/
├── app/
│   ├── layout.js, page.js, globals.css
│   ├── login/, register/, cart/, dashboard/, admin/
│   ├── products/ (listing + [id] detail)
│   └── api/ (auth, products, orders, reviews, chat, cart, recommendations, seed, admin)
├── components/ (AuthProvider, Navbar, ChatbotWidget)
└── lib/
    ├── mongodb.js
    ├── models/ (User, Product, Order, Review, Chat)
    └── ai/ (sentiment.js, recommendations.js, chatbot.js)
```

How to Run

1. Set `MONGODB_URI` in `.env.local` to your MongoDB Atlas connection string
2. `npm run dev` → visit `http://localhost:3000/api/seed` to load 194 products
3. Demo: `user@shop.com` / `user123` · Admin: `admin@shop.com` / `admin123`